Comparison of Anthropometry, Bioelectrical Impedance, and Dual-energy X-ray Absorptiometry for Body Composition in Cirrhosis

Indu Grover *, Namrata Singh *, Deepak Gunjan *, Ravindra M. Pandey †, Hem Chandra Sati †, Anoop Saraya *

*Department of Gastroenterology and Human Nutrition Unit, All India Institute of Medical Sciences, N. Delhi, India and †Department of Biostatistics, All India Institute of Medical Sciences, N. Delhi, India

Background & aims: **This study was planned to evaluate triceps skinfold thickness (TSFT), mid-arm muscle circumference (MAMC) and bioelectrical impedance analysis (BIA) for assessing body composition using dual-energy X-ray absorptiometry (DEXA) (reference) and to predict fat mass (FM) and fat-free mass (FFM) in patients with cirrhosis.** ***Methods:*** **FM and FFM were assessed by using DEXA and BIA. Skin-fold calliper was used for measuring TSFT, and MAMC was calculated. Bland–Altman plot was used to determine agreement and linear regression analysis for obtaining equations to predict FM and FFM.** ***Results:*** **Patients with cirrhosis (n = 302, 241 male, age 43.7 ± 12.0 years) were included. Bland–Altman plot showed very good agreement between BIA and DEXA for the estimation of FM and FFM. Majority of patients were within the limit of agreement: FM (98%) and FFM (96.4%). BIA shows a positive correlation with DEXA:FM (r = 0.73, $P \leq 0.001$) and FFM (r = 0.86, $P \leq 0.001$). DEXA (FM and FFM) shows a positive correlation with TSFT (r = 0.69, $P \leq 0.01$) and MAMC (r = 0.61, $P \leq 0.01$). The mean difference between the observed and predicted value of FM and FFM by BIA in the developmental set was 0.01 and 0.05, respectively; whereas in the validation set, it was −0.13 and 0.86, respectively. The mean difference between the observed and predicted value of TSFT and MAMC in the developmental set was 0.43 and 0.07; whereas, in the validation set, it was 0.16 and 0.48, respectively.** ***Conclusion:*** **Anthropometry (TSFT and MAMC) and BIA are simple and easy to use and can be a substitute of DEXA for FM and FFM assessment in routine clinical settings in patients with cirrhosis. (J Clin Exp Hepatol 2022;12:467–474)**

Cirrhosis of Liver

Malnutrition is common and has been shown to adversely affect the clinical outcome in cirrhosis.[1,2] Its prevalence varies from 60 to 90% and increases as the stage of liver disease advances.[3–5] Malnutrition leads to increased risk of complications, infections, recurrent hospitalizations, poor quality of life, and decreased survival.[6–9] Despite its relevance, malnutrition is frequently underdiagnosed in patients with cirrhosis.[10] Diagnosing malnutrition in liver cirrhosis is equally challenging due to ascites and edema. Hence, nutritional assessment is vital for the early diagnosis of malnutrition, for early intervention to prevent further complications, and to monitor impact of management in patients with cirrhosis.

Body composition analysis is commonly used for nutritional assessment in the early stage and monitoring various diseases, and there are several tools available for its assessment.[11] Body mass index (BMI) is not a good tool for nutritional assessment due to its fallacy in liver disease. Dual-energy X-ray absorptiometry (DEXA) is the reference standard for nutritional assessment because its ability to estimate bone mineral content (BMC), fat mass (FM), and lean body mass (LBM) accurately and quickly.[11] However, due to its unavailability, high cost, and use of ionizing radiation, DEXA is used mainly in research settings for nutritional assessment.

Bioelectrical impedance analysis (BIA), triceps skinfold thickness (TSFT), and mid-arm muscle circumference (MAMC) are relatively inexpensive, noninvasive, and simple techniques for body composition assessment.[12,13] Anthropometric measurements underestimate the prevalence of malnutrition in liver cirrhosis, particularly more so in the early phase of the disease.[14] BIA estimates the FM and FFM through the resistance of the body to a low voltage, high-frequency alternating current, but BIA equations developed in one population is not applicable to different population.[12,13,15] TSFT is simple, easy-to-do technique but with subjective variation in measurements.

Keywords: anthropometric measurements, bioelectrical impedance analysis, cirrhosis, dual-energy X-ray absorptiometry, nutritional assessment
Received: 26.2.2021; *Accepted:* 29.5.2021; *Available online 9 June 2021*
Address for correspondence: Anoop Saraya, Department of Gastroenterology and Human Nutrition Unit, All India Institute of Medical Sciences, N. Delhi, India.
E-mail: ansaraya@yahoo.com
Abbreviations: ANA: anti-nuclear antibody; anti-HCV: anti-hepatitis C virus; anti-LKM1: anti-liver kidney microsomal antibody type 1; ALP: alkaline phosphatise; ALT: alanine aminotransferase; ASMA: anti-smooth muscle antibody; AST: aspartate aminotransferase; BIA: bioelectrical impedance analysis; BMC: bone mineral content; BMI: body mass index; CTP: Child–Turcotte–Pugh score; DEXA: dual-energy X-ray absorptiometry; FM: fat mass; FFM: fat-free mass; HBsAg: hepatitis B surface antigen; MAMC: mid-arm muscle circumference; TSFT: triceps skinfold thickness
https://doi.org/10.1016/j.jceh.2021.05.012

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The anthropometric and BIA methods can give fallacious estimation in patients with decompensated cirrhosis due to ascites, edema, and low muscle mass.[11,12,15]

Few studies directly compare between the various methods for body composition analysis, and these studies are limited by small numbers of patients. There is a good agreement of body fat percentage measurement by DEXA and skin anthropometry in patients with cirrhosis without fluid overload.[16] However, in cirrhosis, only few studies had compared all three methods for body composition analysis.[17] Most of the studies were conducted in Caucasians. So, our aim was to evaluate the agreement between various body compartments measured by using TSFT, MAMC, and BIA with DEXA (as reference) in patients with cirrhosis. Another aim was to develop a regression equation using BIA and TSFT separately to predict FM and to predict FFM by BIA and MAMC separately as measured by using DEXA.

MATERIAL AND METHODS

The data of this observational study were taken from a part of research project (funded by Indian council of Medical research) in which we studied the levels of vitamin D, micronutrients, and bone health in the patients with cirrhosis attending OPD. Data were taken for only those patients in whom DEXA (gold standard) was available. Patients (n = 302) with cirrhosis between 18 and 65 years of all aetiologies were enrolled from August 2012 to July 2016 at a tertiary care center in North India. Patients were excluded if they were taking vitamin D or multivitamin supplementation ≤4 weeks at the time of enrollment, had European Association for the Study of the Liver (EASL) grade 2 or 3 ascites,[18] malignancy, severe cardiovascular, pulmonary, and renal disease, pregnant or lactating women, known cases of osteoporosis, patients on medications, which can affect bone mineral density such as steroids, bisphosphonates, and anti-epileptics and patients who refuse to participate in the study.

A complete etiological workup was done to assign the cause of cirrhosis and which includes- HBsAg, anti-HCV, anti-nuclear antibody, anti-smooth muscle antibody, anti-liver kidney microsomal antibody type 1, lipid profile, plasma blood sugar, serum ceruloplasmin, copper levels, and whenever required liver biopsy was done. Cirrhosis was diagnosed based on clinical features (history and examination, history of decompensation of underlying chronic liver disease), laboratory parameters and imaging findings (ultrasonography or CT abdomen).[19] Disease severity was assessed by Child–Turcotte–Pugh (CTP) score and model for end-stage liver disease. Written informed consent was obtained from all the patients prior to enrolment in the study. Ethical clearance was obtained from the Institute's Ethics committee (Reference No: IEC/NP-146/2012).

Methods for Body Composition Assessment

DEXA (QDR 4500 Acclaim series, Hologic Inc. Waltham, MA) was used to assess the body composition in supine position after overnight fast. DEXA divided total body weight (TBW) into three compartments: FM, lean mass (LM), and BMC. LM and BMC were merged to get FFM.

BIA was performed by leg-to-leg portable impedance analyzer as per manufacturer instructions (Tanita TBF-215, Japan). Height was measured using the built-in height rod in the instrument. Patients were made to stand on the scale bare foot with minimum clothing, with their hands by their sides. For all patients, 0.5 kg was deducted to account for their clothes. All measurements were made in empty stomach after overnight fast and in standing posture on machine platform. The body weight and impedance were measured, from which FM, percentage FM, and FFM were derived. FFM includes water, muscle, and bone. It cannot give separate value for muscle and bone mineral compartment. The result was expressed as TBW = FM + FFM. Three repeat measurements were recorded, and the mean value was taken as final.

Mid-upper arm circumference (MUAC) was taken at the point midway between the acromion and the lateral epicondyle of elbow joint using a plastic coated nonstretchable measuring tape on the left upper arm or nondominant arm to the nearest value of 0.1cm in the sitting posture with the arm keeping by side. TSFT (measures FM of triceps) was measured at the same position as MUAC by single observer using Harpenden's skin-fold calliper, calibrated to exert constant pressure of 10 g/mm^2 on the left upper arm or nondominant arm to the nearest 0.2 mm in the sitting posture with arm keeping by side. It was measured at least three times, and the average of three was taken as final. MAMC (measures FFM) was calculated by deducting the product of pie (π) and TSFT (cm) from MUAC (cm). All anthropometric measurements were done by single observer, who was blinded to the results of BIA and DEXA but was aware of the underlying diseased condition of the patients.

All these patients were on standard therapy for cirrhosis depending on the clinical conditions and complications, which included prophylactic treatment with beta blockers for high-grade esophageal varices, diuretics, and salt restriction in case of ascites and large volume paracentesis (LVP) along with albumin infusion in case of repeated LVP. Prophylactic treatment with antibiotics was used in case of spontaneous bacterial peritonitis. Patients with hepatic encephalopathy were treated with lactulose and rifaximin, whereas patients with variceal bleed were treated with endoscopic therapy.

Statistical Analysis

Data were analyzed by using statistical software STATA 14.0. Categorical data were expressed as frequency and

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percentage. Quantitative data were expressed as mean ± SD and median (min–max) on the data that followed normal and skewed distribution, respectively. Paired t-test was used to compare the difference and estimated error between two methods. Spearman correlation coefficient was used to check the direction and strength of correlation between two methods. Bland–Altman plot was plotted to show the agreement between two methods (DEXA and BIA). The plotted points were examined to see whether they fell within the limit of agreement (upper and lower limit) or beyond the limit as described by Bland-Altman. Regression equation was developed considering DEXA as a dependent variable and BIA as an independent variable. Further data were randomly divided into a developmental data set (n = 200) and validation data set (n = 102). The developmental data set was used to develop regression equation, which was used on the validation data set to determine the ability of regression equation (correction factor) to predict FM and FFM as obtained by DEXA. Only those patients were taken for the analysis whose body composition was done by both methods (BIA and anthropometric). No imputation was done for missing value. Statistically significance was considered when $P < 0.05$.

RESULTS

Demographics

The study population (Supplementary Figure 1) consisted of 302 patients (241 male, 61 female) with mean age and BMI of 43.7 ± 12 years and 22.7 ± 4.2 kg/m^2, respectively. A total of 201 patients had CTP class A, 90 and 11 patients were from CTP B and C class, respectively (Table 1). The most common cause of cirrhosis was viral (46.7%), followed by NAFLD (26.8%) and alcohol (19.2%). Variceal bleeding

Table 1 Demographic Characteristics of Study Subjects (n = 302).

Demographic variables	Mean ± SD or median (min–max) or frequency (%)
Age (years)	43.71 ± 12.09
Gender (male:female)	241:61
BMI (kg/m^2)	22.74 ± 4.16
CTP class	
A	201 (66.6)
B	90 (29.8)
C	11 (3.6)
Model for end-stage liver disease	10.6 ± 4.3
Etiology	
Hepatitis B and C	141 (46.7)
Alcohol	58 (19.2)
NAFLD	81 (26.8)
Others	22 (7.3)
Hepatic encephalopathy	27 (9.0)
Variceal bleeding	158 (52.3)
Haemoglobin (g/L)	11.44 ± 2.68
Blood urea (mg/dL)	22.5 (3.1–129)
Serum creatinine (mg/dL)	0.8 (0.1–5)
Serum calcium (mg/dL)	8.55 ± 0.90
Serum phosphorus (mg/dL)	3.41 ± 0.77
Bilirubin (mg/dL)	1.25 (0.19–11.1)
Total protein (g/dL)	7.34 ± 0.70
Albumin (g/dL)	3.85 ± 0.76
AST (IU)	47 (12–286)
ALT (IU)	38 (9–219)
ALP (IU)	266 (12–1288)

Abbreviations: CTP: Child–Turcotte–Pugh score; ALT: alanine aminotransferase; AST: aspartate aminotransferase; ALP: alkaline phosphatase; NAFLD: nonalcoholic fatty liver disease.

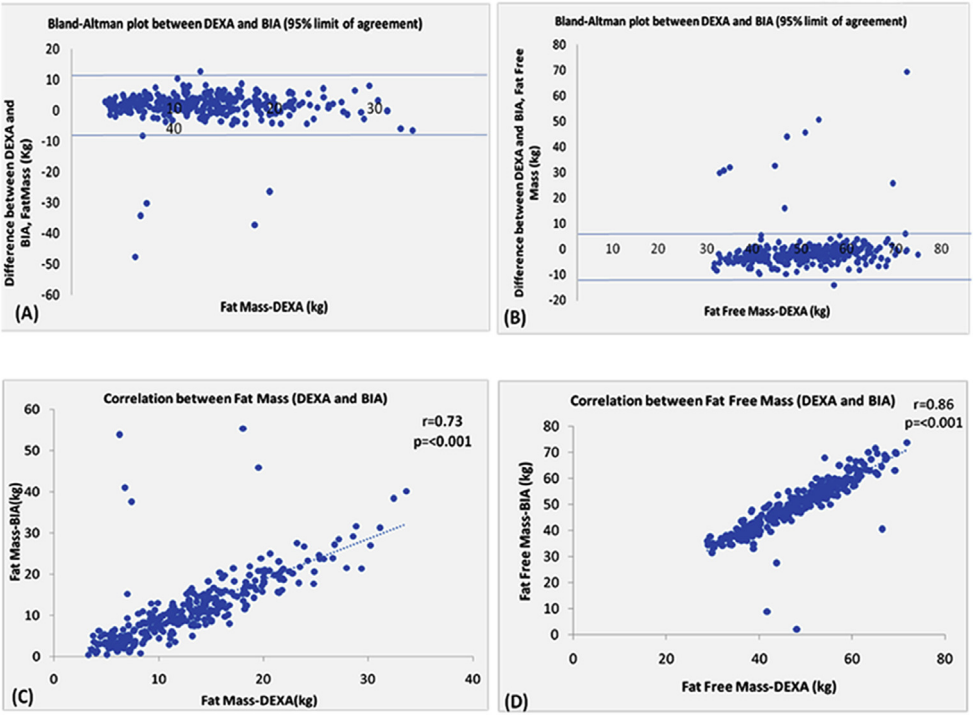

Figure 1 Bland–Altman plot showing the limit of agreement between bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DEXA) for the assessment of fat mass (A) and fat-free mass (B) in the patients with cirrhosis (n = 297). Scatter plot of fat mass (C) and fat-free mass (D) measured using DEXA (kg) and BIA (kg).

was present in 52%, whereas hepatic encephalopathy was present in 9% of patients. Supplementary Table 1 shows body compartment by DEXA scan, whereas Supplementary Table 2 depicted body compositions of different body compartments by BIA and anthropometric measurements by TSFT, MUAC, and MAMC. Supplementary Figure 1 shows consort flow chart for exclusion of the patients in the present study.

Agreement Between Bioelectrical Impedance Analysis and Dual-energy X-ray Absorptiometry

Bland–Altman plot showed very good agreement between BIA and DEXA for the estimation of FM and FFM (Figure 1 A-B). For FM, majority of patients (98%) were within the limit of agreement (−9.7, 12.1). Five patients (1.7%) were out of the upper limit of agreement (+12.1 kg FM) with one patient (0.3%) out of the lower limit of agreement (−9.7 kg FM). Similarly, in FFM, maximum patients (96.4%) were within the limit of agreement (−11.1, 7.8). Ten patients (3.3%) were out of the upper limit of agreement (+7.8 kg FFM) with one patient (0.3%) out of the lower limit of agreement (−11.1 kg FFM).

FM: Body FM measured by using BIA was underestimated (12.09 ± 8.23 vs. 13.17 ± 6.17 kg) compared with DEXA. The difference of FM by DEXA was 1.18 kg (0.54–1.81) (Table 2). DEXA and BIA show a positive correlation in estimating FM (r = 0.73, $P \leq 0.001$) (Figure 1-C). Furthermore, the whole cohort was divided into the developmental and validation set for regression equation between DEXA and BIA methods for FM (Table 3). The difference between the observed and predicted value of FM in the developmental set was 0.01, and in the validation set, it was −0.13. The difference observed between BIA and DEXA in the developmental and validation set shows good correlation between both methods for estimation of FM in patients with cirrhosis (Table 4).

FFM: Body FFM measured by BIA was overestimated (49.62 ± 9.57 vs. 47.86 ± 9.06 kg) compared with DEXA.

Table 2 Limit of Agreement (95% CI) [n = 297], Between Dual-energy X-ray Absorptiometry (DEXA) and Bioelectrical Impedance Analysis (BIA) for Fat Mass and Fat-free Mass.

Body compartments	DEXA (Mean ± SD)	BIA (Mean ± SD)	Difference (95% CI)
Fat mass (kg)	13.17 ± 6.17	12.09 ± 8.23	1.18 (0.54–1.81)
Fat-free mass (kg)	47.86 ± 9.06	49.62 ± 9.57	−1.66 (−2.21, −1.11)

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Table 3 Assessment of Body Composition by Different Methods in the Whole Cohort; and Developmental and Validation Set.

Assessment of body composition	Body compartments	Total mean ± SD (min–max)	Development set (n = 200) Mean ± SD (min–max)	Validation set (n = 102) Mean ± SD (min–max)
DEXA	Fat Mass (kg)	(n = 302) 13.16 ± 6.16 (3.29–33.64)	(n = 200) 13.40 ± 6.12 (3.29–33.64)	(n = 102) 12.69 ± 6.24 (3.58–28.85)
	Fat-free mass (kg)	47.85 ± 9.36 (28.82–71.70)	48.14 ± 9.40 (28.82–71.70)	47.28 ± 9.30 (30.16–69.43)
BIA	Fat mass (kg)	(n = 297) 12.08 ± 8.23 (0.44–55.3)	(n = 196) 12.50 ± 8.66 (0.44–55.3)	(n = 101) 11.65 ± 7.66 (0.5–37.6)
	Fat-free mass (kg)	49.61 ± 9.57 (2.2–73.7)	49.51 ± 10.00 (2.2–73.7)	49.73 ± 8.80 (33.5–69.9)
TSFT (triceps fat mass)	Fat mass (cm)	(n = 297) 1.45 ± 0.77 (0.28–7)	(n = 196) 1.48 ± 0.69 (0.32–3.5)	(n = 101) 1.41 ± 0.90 (0.28–7)
MAMC (mid-upper arm fat-free mass)	Fat-free mass (cm)	(n = 297) 22.29 ± 3.16 (6.16–34.65)	(n = 196) 22.52 ± 3.22 (6.16–34.65)	(n = 101) 21.84 ± 3.00 (7.02–28.35)

Abbreviation: BIA: bioelectrical impedance analysis; DEXA: dual-energy X-ray absorptiometry; MAMC: mid-arm muscle circumference; TSFT: triceps skinfold thickness.

The difference of FFM by DEXA was −1.66 kg (−2.21, −1.11) (Table 2). DEXA and BIA show a positive correlation in estimating FFM (r = 0.86, $P \leq 0.001$) (Figure 1-D). Same developmental and validation set for regression equation between DEXA and BIA for FFM was prepared (Table 3). The difference between the observed and predicted value of FFM in the developmental set was 0.05 and 0.86; in the validation set, it shows good correlation between both methods for estimating FFM in the patients with cirrhosis (Table 4).

After excluding CTP A cirrhosis, the agreement and correlation between DEXA and BIA for FM and FFM in CTP B or C (n = 101) were similar as for the whole group.

Correlation Between Anthropometry and Dual-energy X-ray Absorptiometry

FM: FM measured by DEXA and TSFT shows a positive correlation (r = 0.69, $P \leq 0.01$) (Figure 2-A). The difference in observed; predicted value for FM by DEXA and TSFT in developmental set was 0.43, and in the validation set, the difference was 0.16.

FFM: FFM measured by MAMC shows a positive correlation (r = 0.61, $P \leq 0.01$) (Figure 2-B). The difference was 0.07 and 0.48 in developmental and validation set, respectively, between observed and predicted value for FFM by MAMC against DEXA. So, the correlation between MAMC and DEXA is good for FFM estimation (Table 4).

DISCUSSION

In this study, we compared the utility of BIA and anthropometric methods for assessing FM and FFM with reference method DEXA in patient with cirrhosis. We found that BIA is accurate in estimating FM and FFM in patients with cirrhosis compared with reference standard DEXA in this study. The coefficient of determination (R^2) showed a

Table 4 Regression Equation for the Prediction of Fat Mass, Fat-free Mass, and Agreement in the Developmental and Validation Set.

Regression equation	R^2	Mean difference in observed – predicted (95% CI)	
		Developmental set (n = 200)	Validation set (n = 102)
FM (DEXA) = 7.3 + 0.49 FM (BIA)	0.4946	0.01 (−0.60, 0.62)	−0.13 (−0.89, 0.64)
FM (DEXA) = 3.72 + 6.43 (TSFT)	0.5664	0.43 (−0.12, 0.99)	0.16 (−0.90, 1.22)
Fat-free mass (DEXA) = 10.09 + 0.76 FFM (BIA)	0.6960	0.05 (−0.67, 0.77)	0.86 (0.20, 1.5)
Fat-free mass (DEXA) = 6.11 + 1.86 (MAMC)	0.4124	0.07 (−0.93, 1.1)	0.48 (−1.06, 2.0)

Abbreviation: BIA: bioelectrical impedance analysis; DEXA: dual-energy X-ray absorptiometry; FM: fat mass; MAMC: mid-arm muscle circumference; TSFT: triceps skinfold thickness.

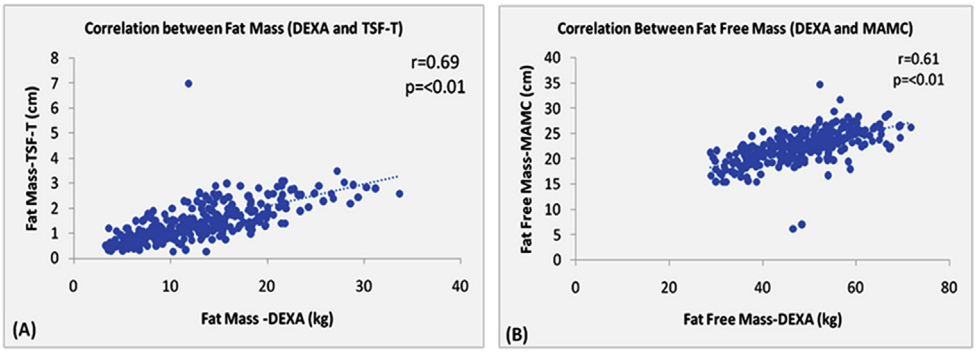

Figure 2 Scatter plot of fat mass (A) measured using dual-energy X-ray absorptiometry (DEXA [kg]) and triceps skinfold thickness (TSFT [cm]); and scatter plot of fat-free mass (B) measured using DEXA (kg) and mid-arm muscle circumference (MAMC [cm]) (n = 297).

better agreement between DEXA and BIA for FFM ($R^2 = 0.69$) than for FM ($R^2 = 0.49$). So, DEXA may be replaced with BIA to estimate FFM as well as FM in patients with cirrhosis. BIA is found to be reliable tool to assess malnutrition in 41 cirrhotic patients with and without ascites;[20] however, in another small study, BIA was not found to be reliable for nutritional assessment in patients with cirrhotic ascites.[21] Another large study estimated that BIA and skin anthropometry are poor tools for measuring body composition in patients with cirrhosis, compared with DEXA.[22] BIA is simple and rapid and can be used bedside. Both DEXA and BIA are accurate in measuring FM and FFM in normal hydration status. However, several factors affect BIA measurements such as fluid overload, age, gender, body characteristics like amputation, and race.[13,15,23] Phase angle measured by vectorial analysis is said to be less affected by overhydration and a better parameter to estimate nutritional status in liver cirrhosis.[24,25] Phase angle characterizes both the amount and quality of soft tissue, which represents cellularity and cell functions. The higher the phase angle is, better would be the health outcome and lower the phase angle worse would be the clinical outcomes.[24,25] Body composition changes nonlinearly on hydration when measured with both BIA and DEXA.[26] Our study results are at variance with other studies may be due to including both patients with and without ascites. Our study group is also racially different as most of the studies are conducted in Caucasian populations.[14,16,17]

Anthropometric methods, TSFT for FM and MAMC for FFM less accurately estimate FM and FFM in comparison with DEXA, whereas BIA has more precise estimate for FM and FFM in patients with cirrhosis. The coefficient of determination (R^2) showed a closer agreement for FM by TSFT ($R^2 = 0.56$) and FFM by MAMC ($R^2 = 0.41$). A study by Morgan et al. suggested that two-compartment model is of limited value for the body composition assessment in patients with cirrhosis, which is in accordance with our study.[27] In another study, percentage body fat estimated by skin anthropometry was similar to DEXA in patients with cirrhosis without fluid overload.[16] Both these studies included small number of patients and without fluid overload.

In our study, only one site (TSFT) skin anthropometry was used to estimate body FM. Skin anthropometry gives only the estimate of subcutaneous fat, whereas visceral fat is another major component of body fat, which cannot be estimated by skin anthropometry, and there are ethnic differences in body fat distribution in Southeast Asians.[28] We felt that probably measurements at four sites for skin anthropometry would have given better estimates and correlation with DEXA for FM. Anthropometric measurements can be performed in both inpatient and outpatient care as a bedside tool and can be done serially. Anthropometric measurements are simple, rapid, noninvasive, and cost-effective too. Anthropometric measurements can give fallacious results due to interobserver variations, and there is no established cutoffs in cirrhotic patients.[23] In this study, only single observer took all anthropometric measurements; hence, only interobserver bias cannot account for this difference.

Body composition analysis is assessed by various methods, but DEXA is regarded as the reference standard.[11] FM provides the estimate of the overall fat compartment, and it can be measured with different methods, whereas LBM encompasses muscle and water. DEXA measures three body compartments (bone mineral, LBM, and FM), whereas BIA and anthropometry measures only two compartments (FM and FFM).[11] Apart from risk of ionizing radiation, cost, immobility, and availability of DEXA is prohibitive for common use in clinical practice. DEXA also cannot give good estimate of water content, and repeated measurement for monitoring and follow-up in routine clinical practice is difficult. However, DEXA is the commonly used method to validate the body composition analysis by other methods, due to its reproducibility, accuracy, and better characterization of LBM and BMC.[11] This is the reason we did not compare between two- and three-compartment methods for LBM.

We included a large number of patients with cirrhosis of mixed etiology, and measurements were validated internally. These results should be replicated in other cohorts for external validation. Limitations of our study that can accrue in the patients with cirrhosis with ascites, can increase the

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fallacy of measurements; however, we tried to minimize that by excluding EASL grade 2 or 3 ascites and excluding very sick patients. Another limitation was that we did not measure skinfold thickness at four sites, and there was no control group in our study, which would have given better insight into the comparison between different methods. All the BIA measurements were done by single-frequency BIA, which measures only two body compartments, do not provide value of phase angle, and provide less accurate estimation of FM and FFM, compared with multifrequency BIA.

In conclusion, BIA has good agreement with DEXA for assessment of FM and FFM in patients with cirrhosis. BIA and anthropometric measurements can be used for FM and FFM assessment in routine clinical setting in patients with cirrhosis. However, for an accurate and precise estimate of body composition, DEXA is still an invaluable but expensive tool.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Indu Grover: Conceptualization, Data curation, formal analysis, Investigation, Methodology, Visualization, Writing - original draft; **Namrata Singh**: Formal analysis, Investigation, Methodology, Supervision, Validation, Writing - original draft, Writing - review & editing; **Deepak Gunjan**: Formal analysis, Investigation, Supervision, Writing - review & editing; **R. M. Pandey**: Formal analysis, Software, Supervision, Writing - review & editing; **Hem Candra Sati**: Formal analysis, Software; **Anoop Ssaraya**: Conceptualization, formal analysis, Funding acquisition, Project administration, Validation, Visualization, Writing - review & editing.

CONFLICTS OF INTEREST

The authors have none to declare.

ACKNOWLEDGEMENT

This work was financially supported by Indian Council of Medical Research (ICMR), New Delhi, India. The authors are grateful to Dr. Ravinder Goswami, Professor, Department of Endocrinology and Metabolism, All India Institute of Medical Sciences, New Delhi for allowing us to conduct DEXA scan in his department.

FUNDING

None.

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SUPPLEMENTARY DATA

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jceh.2021.05.012.

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